Case 13: WUI Parametric Response Maps

1 Verification purpose

This case defines a parametric verification of the resolution-independent WUI response to generation rate, inter-structure separation, and wind speed. The two controlled response families contain 270 requested points with duplicate baselines removed.

The response test is valid only when the integrated-structure capability is available. The current source gates that capability, so this case verifies design completeness and reports the limitation explicitly instead of substituting the native cell-local model.

2 Mathematical formulation and numerical implementation

Each physical structure is a square of side $a = 10\,\mathrm{m}$. Although the computational grid uses $\Delta x = 2.5\,\mathrm{m}$, all cells belonging to one structure must act as one source and one collector. For physical structure s, the required formulation is

$$\mathrm{HRR}_s(t) = q''(t)a^2, \qquad \dot{N}_s(t) = GR \times \mathrm{HRR}_s(t), \qquad N_s(t) = \sum_{j \in s} N_j(t).$$

Transport is evaluated once for the structure-scale heat-release rate (HRR), deposition is accumulated over its physical footprint, and the resulting load N_s/a^2 drives the physical ignition criterion. This grouping is the key difference from the native multiple-discrete-cell treatment, in which one physical structure is fragmented into independently emitting and igniting raster cells.

The level-set time step is

$$\Delta t = 0.5\frac{\Delta x}{U_{\mathrm{wind}}},$$

and surface spread is disabled. Spread success therefore means that the firebrand process alone propagates through the structural array. For each parameter point, mean rate of spread (ROS) is obtained by fitting structure position against first-ignition time.

The first family varies

$$d = 2.5, 5, \ldots, 30\ \mathrm{m}, \qquad GR = 5, 10, \ldots, 50\ \mathrm{pcs\,MW^{-1}\,s^{-1}},$$

with a fixed 40 mph wind. The second fixes the separation at 10 m, uses the same generation rates, and varies wind speed from 10 to 150 mph in 10 mph increments.

3 Derivation of expected behavior

Increasing separation moves targets into lower-probability transport tails, while increasing generation raises deposited load. Wind produces competing reach and ignition-cooling effects, so the critical generation boundary should rise at both low and high wind speeds.

The design count follows directly from the generated parameter products:

$$12\ \text{separations} \times 10\ GR = 120, \qquad 15\ \text{winds} \times 10\ GR = 150, \qquad N_{\mathrm{design}} = 120 + 150 = 270.$$

The two baseline-equivalent coordinates remain distinct members of their respective response families. At the design-fire plateau, a physical structure has $HRR_{\mathrm{max}} = 400(100) = 40$ MW; consequently the

tested generation range corresponds to

$$\dot{N}_{\max} = GR(40) = 200,\ 400, \ldots, 2000 \text{ pcs s}^{-1}.$$

With $\Delta x = 2.5$ m and $CFL = 0.5$, the generated level-set step is $0.5(2.5)/[U_{\text{mph}}(0.44704)]$: it is 0.069904 s at 40 mph, 0.279617 s at 10 mph, and 0.018641 s at 150 mph.

For each coordinate, GR_c is selected from $\{5, 10, \ldots, 50\}$; if no point propagates, the rank-only sentinel is the next design level, 55. Thus a one-level wind-edge excess is $55-50 = 5$ pcs MW^{-1} s^{-1}, exactly the acceptance increment. The moderate set is 30–120 mph and the edge set is $\{10, 20, 130, 140, 150\}$ mph. The explicit wind metrics are

$$\overline{GR}_{c,\text{edge}} - \text{median}(GR_{c,\text{moderate}}) \geq 5, \qquad \frac{\overline{ROS}_{\text{moderate}}}{\overline{ROS}_{\text{edge}}} \geq 1.10.$$

The separation expectation is strong positive rank association of d with GR_c, negative rank association of d with ROS, and positive rank association of GR with ROS. These become $\rho \geq 0.85$, median $\rho \leq -0.50$, and median $\rho \geq 0.70$, respectively, with the independent response-scale ceiling max $ROS \leq 0.08$ m s^{-1}.

4 Verification metrics and acceptance criteria

For a coordinate value z, define the critical generation rate

$$GR_c(z) = \min\{GR : \text{the array sustains downwind spread}\}.$$

When no tested value succeeds, the postprocessor assigns the next design level, 55 pcs MW^{-1} s^{-1}, only for rank and boundary metrics; it does not reinterpret failure as an observed threshold.

For the separation family, the documented response predicts a nonlinear increase of GR_c with d, from approximately 2.5 to more than 30 pcs MW^{-1} s^{-1} [1, 2]. ROS should decrease with separation and increase with generation. The verification therefore requires Spearman rank correlation $\rho(d, GR_c) \geq 0.85$, median $\rho(d, ROS) \leq -0.50$, and median $\rho(GR, ROS) \geq 0.70$.

For the wind family, the critical boundary is U-shaped. Low wind gives insufficient reach and a high ignition threshold; very high wind improves transport but suppresses the smoldering-to-flaming transition through convective cooling. The mean critical generation rate outside 30–120 mph must exceed the median within that interval by at least 5 pcs MW^{-1} s^{-1}. Mean ROS in the moderate range must be at least 1.10 times the edge-wind mean. Finally, firebrand-driven ROS must remain below 0.08 m s^{-1}, consistent with the documented response scale [1, 2].

4.1 Justification of metric quantification

The rank limits quantify robust trends without assuming that the nonlinear responses are linear. A separation-threshold Spearman coefficient of 0.85 requires strong ordering across twelve separation levels while allowing a small number of adjacent ties or reversals near a discrete ignition boundary. The -0.50 ROS/separation and 0.70 ROS/generation limits require, respectively, at least a moderate negative and a strong positive monotone association despite flat-response regions. They therefore test the direction and persistence of the mechanisms rather than a fitted slope chosen for one parameter interval.

The 5 pcs MW^{-1} s^{-1} wind-edge excess equals one full generation-rate design increment, so the U-shaped boundary must be resolved by the experiment rather than inferred from sub-grid interpolation. The 1.10 moderate/edge ROS ratio requires a 10% effect beyond equality. The 0.08 m s^{-1} ceiling is a

scale check taken from the reported firebrand-driven response range for this design [1, 2]. All 270 points and the resolution-independent structure capability are mandatory because missing edge or threshold points can materially change both rank statistics and the inferred U shape.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Design completeness	Unique generation–separation and generation–wind points after baseline deduplication.	270 of 270 points
Required capability	Resolution-independent single-structure source, collector, and input data conversion procedure.	Available
Result completeness	Current admissible result rows divided by 270.	100%
Threshold/separation rank	Spearman $\rho(d, GR_c)$.	≥ 0.85
ROS/separation rank	Median Spearman $\rho(d, ROS)$ across generation levels.	≤ -0.50
ROS/generation rank	Median Spearman $\rho(GR, ROS)$ across both response families.	≥ 0.70
Wind-edge threshold excess	Mean critical generation outside 30–120 mph minus the median inside that range.	$\geq 5\,\mathrm{pcs\,MW^{-1}\,s^{-1}}$
Moderate/edge ROS ratio	Mean ROS at 30–120 mph divided by mean ROS outside that range.	≥ 1.10
Maximum ROS	Maximum firebrand-driven ROS over all admissible design points.	$\leq 0.08\,\mathrm{m\,s^{-1}}$
Overall decision	Capability, complete results, and every trend/scale metric are conjunctive.	PASS only if all pass

5 Simulation configuration and predicted outcome

Although execution remains capability-gated, every design point now carries an auditable spatial configuration. Structures are rasterized as a periodic two-dimensional lattice over a 220×100-m physical domain. The generation–separation family applies its selected gap in both coordinate directions; the generation–wind family uses the 10 m baseline gap. Every footprint in the first upwind structure column is marked by the non-positive initial level-set field.

Figure 1 shows the baseline-separation geometry. The two-cell numerical halo is excluded. The left panel verifies repetition in both directions, while the right panel verifies that ignition spans the entire first structure column instead of selecting only the lower-left structure.

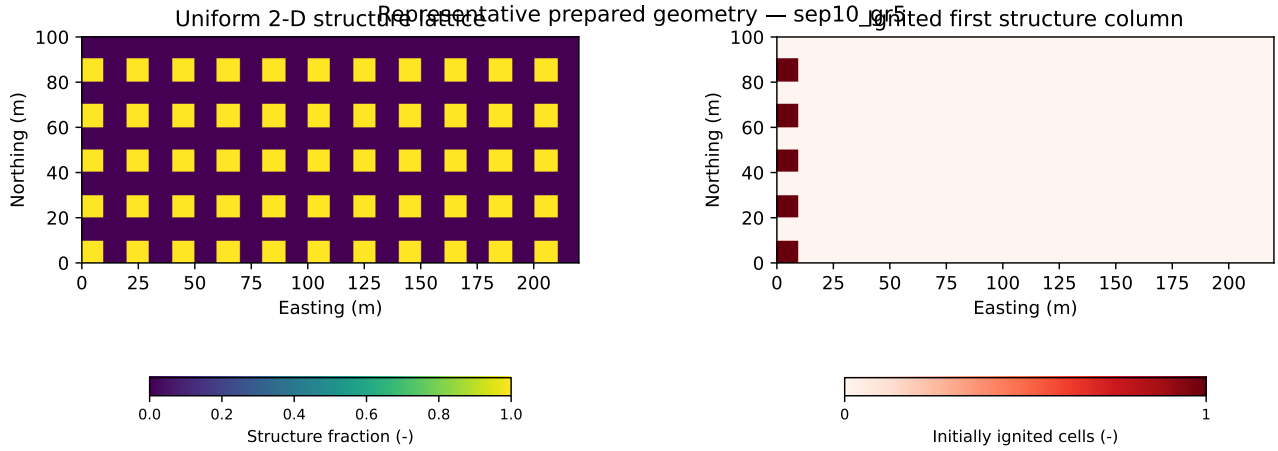


Figure 1: Representative prepared two-dimensional structure lattice and first-column initial ignition mask for the capability-gated parametric design.

The Courant–Friedrichs–Lewy (CFL) number measures the distance advected during one timestep relative to the grid spacing and constrains the time discretization.

Table 2: Controlled parametric design.

Parameter	Separation family	Wind family
Grid spacing	2.5 m	2.5 m
Physical domain	220 × 100 m	220 × 100 m
Structure side	10 m	10 m
Separation	2.5–30 m, step 2.5 m	10 m
Wind	40 mph	10–150 mph, step 10 mph
Generation	5–50, step 5	5–50, step 5
Spatial treatment	single physical unit	single physical unit
Initial ignition	complete first structure column	complete first structure column
CFL based on wind	0.5	0.5
Surface spread	disabled	disabled
Number of points	120	150

Expected ignition classifications form an increasing boundary in the (d, GR) plane and a U-shaped boundary in the (U_{wind}, GR) plane. The ROS maps should be largest for high production, short separation, and moderate wind. Hamada-model curves included in the documented parameter comparison [1, 2] are contextual because Hamada includes radiation and flame contact; they are not acceptance references for this firebrand-only test.

5.1 Preprocessing, execution, and postprocessing

The parameter design contains all 270 prescribed combinations. Each defines co-registered structure-fraction, structure-identity, fuel-model, and initial level-set fields. A combination is simulated only if the tested ELMFIRE version supports the resolution-independent structure formulation and its required input representation. Substituting the native cell-based formulation would test a different discretization and would not establish the intended result.

Analysis requires the specified quantities for the complete parameter design, derives critical boundaries, calculates the rank and response metrics, and constructs the classification and spread-rate compar-

isons. Missing evidence remains NOT EVALUABLE; an unavailable model capability is not replaced by a different physical or numerical formulation.

6 Actual simulation results

No whole-domain ELMFIRE result is available because the required integrated-structure capability is absent and no simulation condition is runnable under the current source interface. The postprocessor therefore does not manufacture a raster or substitute an analytical field for model output. The case remains NOT EVALUABLE, and the response map below is reported only as the prescribed verification envelope.

0 of 270 exact-design result points are present. The decision is **NOT EVALUABLE**. Figure 2 shows only the expected qualitative boundaries while the capability is absent. Every panel is labeled accordingly; no guide curve is represented as an ELMFIRE result.

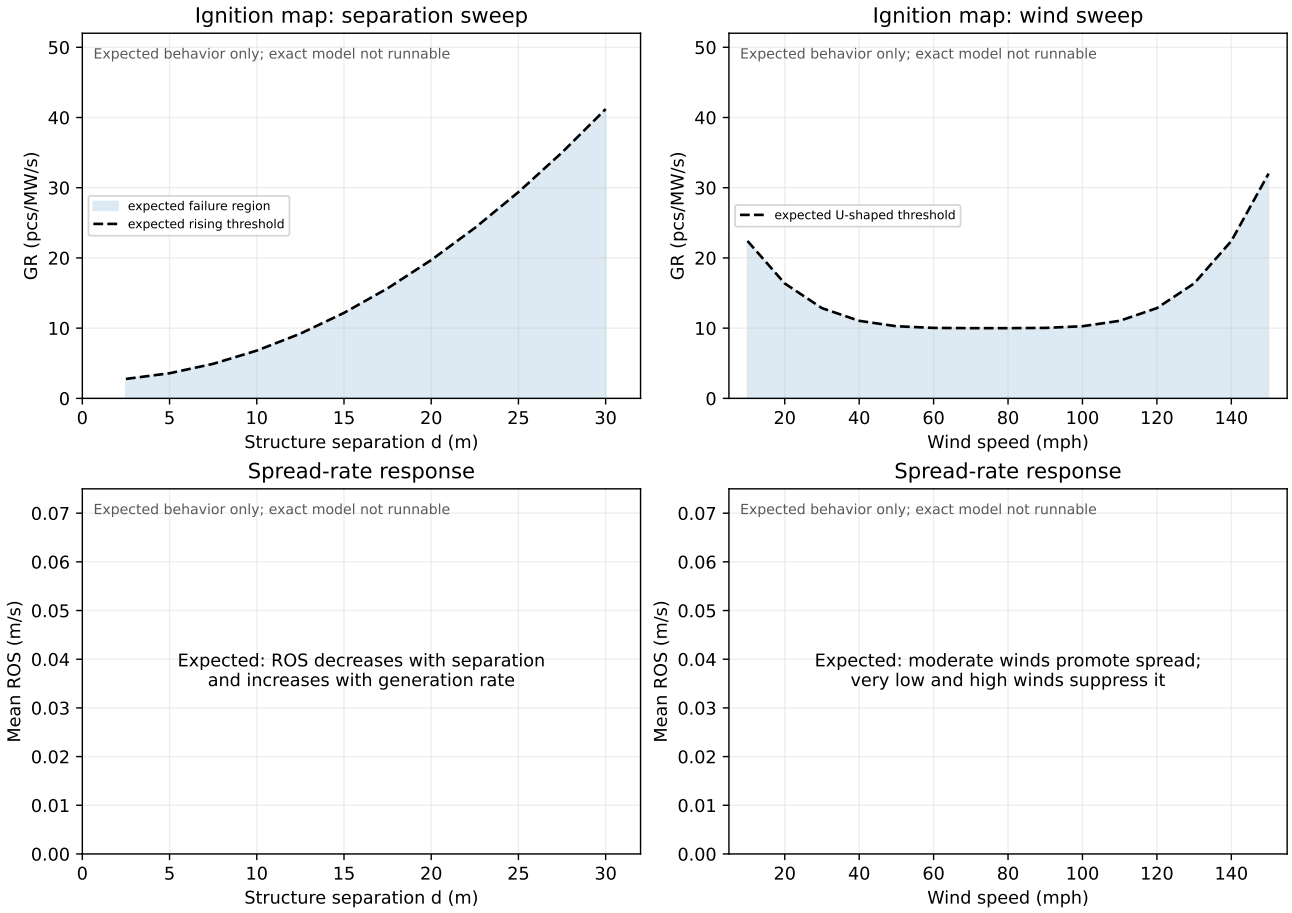


Figure 2: Parametric-response verification layout. The upper panels reproduce the coordinate systems of the two parametric reference maps; the lower panels are reserved for the corresponding ROS curves. In the current capability-gated state, shaded regions and dashed lines communicate expected behavior only.

7 Calculated metrics and verification decision

Table 3: Calculated verification metrics and component decisions.

Metric	Acceptance limit	Calculated	Status
Parameter-design completeness	270/270	270	PASS
Exact single-structure capability	available	N/A	NOT EVALUABLE
Current result completeness	100%	0	NOT EVALUABLE
Critical GR versus separation Spearman rho	≥ 0.85	N/A	NOT EVALUABLE
ROS versus separation median Spearman rho	$\leq$ -0.50	N/A	NOT EVALUABLE
ROS versus generation median Spearman rho	≥ 0.70	N/A	NOT EVALUABLE
Low/high-wind threshold excess	≥ 5 pcs/MW/s	N/A	NOT EVALUABLE
Moderate/edge-wind ROS ratio	≥ 1.10	N/A	NOT EVALUABLE
Maximum firebrand-driven ROS	≤ 0.08 m/s	N/A	NOT EVALUABLE

The current decision is **NOT EVALUABLE**, not a failure of the firebrand equations. The intended resolution-independent parameter sweep cannot be executed because the current source does not implement the single-structure source and collector required by the cited model description [1]. The native model would split each 10-m structure into sixteen 2.5-m cells and change HRR, transport, and accumulation; those outputs are not admissible evidence for this case.

The complete parameter design, expected behavior, result configuration definition, metrics, figure generator, and standalone report are now present. Once the source feature and its input data conversion procedure are implemented, the capability gate should be enabled, all 270 points executed, and the same postprocessor used without changing acceptance limits.

8 Technical interpretation

The capability gate prevents cell-local structural discretization from being presented as evidence for a single-structure parametric model. Once admissible results exist, boundary and rank metrics distinguish physical response trends from isolated successful points.

9 Scope and limitations

The design covers idealized, uniform arrays and verifies qualitative model response over the declared parameter grid. It remains not evaluable until the required single-structure source and collector are implemented.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.

[2] Y. Qin et al., "Simulations of firebrand-driven fire spread in landscape-scale Wildland–Urban-Interface and urban conflagration models," *Fire Safety Journal*, vol. 162, 104686, 2026, doi:10.1016/j.firesaf.2026.104686.